# Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

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Paper ID *****

## Abstract

*Replaying a few hundred stored documents comes close to joint training on average when a continual key-information extractor meets new domains, yet fails when its label space grows. We ask why, as an investigation rather than a method. On a class-incremental stream of receipts we test a ladder of hypotheses, each with its prediction stated in advance, and measure where the errors go. Forgetting is not simply harder: with a whole-document memory and a shared softmax the model barely learns the last sessions it is trained on, which naive training learns well. Stale stored labels explain part of it: giving the older stored documents their true labels at each boundary leaves the model below joint training (three seeds so far). Stopping the sharing of the background class closes most of the gap, as existing per-session readouts do: with the same memory and no true labels, one background logit per session reaches* 89.1 *full-label F1 against* 92.0 *for joint training. The principle, familiar from training on partially annotated data, is that an "O" label is evidence only against the classes its annotator could see. Where label growth supplies the routing key, on a single-schema stream, per-session low-rank trunk updates can replace the memory; across schemas they did not, in one run.*

## 1. Introduction

Key-information extraction (KIE) labels each OCR token of a visually-rich document (form, invoice, receipt), using its words, its 2-D layout and the page image; multimodal encoders such as LayoutLMv3 [18], LiLT [50] and BROS [16] define the state of the art. Deployed extractors meet new document types and growing schemas, while retraining on all history is blocked by cost and by privacy constraints: a continual-learning (CL) problem, whose central obstacle is catastrophic forgetting [33].

Replay is empirically the strongest remedy in the CL toolbox [32, 51], and on documents it looks sufficient. On LayoutLMv3, a stream of forms and receipts under one fixed label space is learned by plain replay (ER) with 200 stored documents to 87.9 AA, within 3.5 points of joint training on every seed (mean 88.7; App. F). Let the label space grow instead, by adding CORD's 30 receipt fields six at a time, and the same method (200 documents) reaches 24.3 AA, against 91.3; both ER cells are legacy-protocol runs (†) that early-stop on the test split, which favours ER (Sec. 3). Distillation, prompts, low-rank adapters and the continual-NER methods built for the shift of the "O" label do no better (Tab. 3). A class-incremental stream is expected to be harder; what is not obvious is why a memory that holds the old documents does not help. This paper is about that question: *why does replay, which solves domain shift, fail when the label space grows?*

We treat it as an investigation (Tab. 1): we write down the explanations a reader would offer, state what each predicts before its test, test each with an intervention on the same encoder, stream and recipe, and measure where the final model's errors go (Fig. 2). Forgetting is not just harder (H0): with a whole-document memory that retains under domain shift, the model suppresses the last sessions it is trained on, which naive training does not do. Stale labels in the memory (H1) are part of the cause: relabelling the older

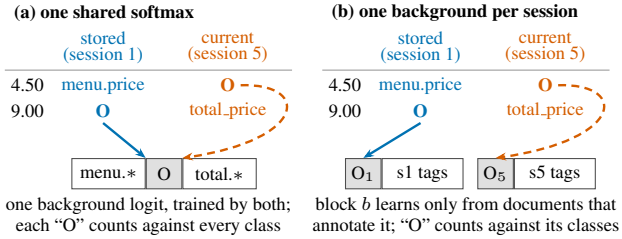



Figure 1. **The background conflict on one pair of price tokens.** A receipt stored in session 1 labels its total "O"; a current receipt of session 5 labels its item price "O". (a) With one shared softmax both "O" labels train the same background logit, against every class. (b) With one background per session (blocks 1 and 5 shown, each with its session's tags), an "O" is evidence only against the classes its annotator could see.

stored documents at each boundary recovers much of the gap but stays below joint training and below the same memory with a per-session background (three seeds). Whether what current documents say about old entities is the rest (H2) is open: label-side repairs we ran earlier are preliminary, and a MiB-style unbiased loss on the same memory is the decisive test. Removing the shared background (H3, Fig. 1) closes most of the gap with no true labels, given a memory, and then per-session isolation can replace the memory (H4), but only where label growth supplies a routing key.

None of the ingredients is new. The background shift is known from class-incremental segmentation [5, 7, 10], from incremental detection, where both its stored side [28] and its two-sided form [35] have been named, and from continual NER [30, 36, 56]; separated softmaxes [1], per-task heads with a "none of mine" output [20] and per-session softmaxes over a constant Other logit [27] already undo the shared normalisation; forgetting in the output-facing layers is known from image classifiers [40]. What has not been done is to take apart, on token-labelled documents, why replay fails: which part a memory repairs, which part survives a memory with true labels, and what the errors of each look like.

**Contributions.**

1. **The investigation.** A decomposition of why replay fails under label growth that prior work did not run on token-labelled documents: a memory $\times$ readout factorial, the stored-label side reduced by a relabelled memory with the gap that remains, and measurements of where the errors go under each readout (Sec. 4): with a shared softmax old entities are missed at "O" with or without a memory (by a span-level indicator), and a memory adds a suppressed current session; with a blocked readout and no memory, drift sends them to another session's class through over-claiming blocks. The relabelled-memory control rests on three seeds so far.
2. **An empirical finding, not a new mechanism.** A per-session background readout, as in ICE-O and MORE, combined with whole-document replay and a supervision rule in the spirit of training on disjoint tag sets, closes most of the gap to joint training in continual KIE on every seed, where the continual-NER and replay methods in our comparison do not. Our instantiation (SBR) differs from ICE-O in a learned background per session and in old blocks kept trainable from the memory (Sec. 5).
3. **Boundary conditions.** With the conflict removed, per-session low-rank trunk deltas (SBR-Recall) replace the memory on a single-schema stream, within 2.1 points of joint training on every seed under both metrics, and fail without a memory across annotation schemas (one seed), where the task-identity view says they must (Sec. 5.3).

## 2. Related Work

**Background shift.** When each step annotates only its own classes, the background label changes meaning. In class-incremental segmentation MiB [5] matches a background label with the probability of background or any old class in the loss, and the old model's background with background or any new class in the distillation; PLOP [10] and SSUL [7] pseudo-label the background with the old model, and SSUL also freezes the backbone and the past classifiers, keeps a tiny memory and uses per-class sigmoids. Incremental detection has named both its stored side [28] and its two-sided form [35]. In continual NER the shift of "O" is the unlabelled-entity problem [30], answered by distillation on "O" tokens [36], pseudo-labels [56] and other repairs (App. J). We do not claim the conflict; we separate its stored and current sides with a relabelled memory and measure what each leaves.

**Separated readouts and task identity.** A softmax over old and new classes together biases them, and separating it removes the bias [1]; ER-ACE [4] masks the old classes out of the softmax of incoming samples only. Giving each task its own "none of mine" output turns class-incremental learning into within-task prediction plus task identification [21]: MORE [20] isolates each task's parameters with hard-attention masks on adapters of a frozen pre-trained transformer, trains a head per task whose out-of-distribution output learns from saved samples, and lets the most confident head answer; in class-incremental extraction ICE-O [27] trains, on a frozen encoder and with no memory, one softmax per session over a constant Other logit and that session's types, so that a mention is Other only if every session scores below that logit. SBR is this pattern on documents with a learned background per session and old blocks kept trainable from a memory. SBR-Recall shares MORE's isolation, with per-session low-rank trunk deltas, but trains each block's "none of mine" logit on its own session's negatives, with no saved samples. Adapter merging and disjoint-tag-set taggers: App. J.

**Continual document IE.** Document-level continual learning is recent and small: multimodal LLMs forget under fine-tuning [55], and GDI-Bench [24] limits it by freezing most parameters; continual table detection [34] uses plain ER; ProtoNER [23] adds key-information classes from a few documents with prototypes; AMD-Proj [44] projects gradients against a memory of task subspaces; and Sassioui et al. [45] benchmark layout-aware models by stream regime, finding that no method they test consistently preserves old knowledge under label-space shift and that, for the non-replay methods they test, domain transfer forgets most. They decompose forgetting by regime of the stream; we decom-

Table 1. **The hypothesis ladder** (CIL-CORD, LayoutLMv3 unless stated; mean ± s.d. over three seeds, per-seed values when fewer; $^{n=k}$: $k$ seeds so far; $^{\dagger}$legacy protocol; "–": no full-label run). Verdict: what the row says about its rung's prediction, on every seed; "open": only preliminary, cross-protocol evidence so far. $^{r}$At each boundary the documents stored before the session just finished receive their true labels (Sec. 4.2). $^{e}$ER also trains every layer and keeps 200 raw documents, so it changes more than the memory format (Sec. 4.1).

| Rung | Hypothesis | Prediction if true | Intervention (CIL-CORD unless stated) | Full-label F1 | AA | Verdict |
|---|---|---|---|---|---|---|
| | | | Naive fine-tuning (reference) | 13.0±0.2 | 18.3±0.5 | |
| H0 | Only drift: forgetting is just harder | a memory that retains under domain shift retains here; raw beats compressed | whole-document memory (CoLaR), shared softmax | 30.7±1.5 | 33.5±1.0 | refutes |
| | | | raw-document memory (ER), shared softmax$^{e}$ | – | 24.3±2.4$^{\dagger}$ | refutes |
| H1 | Stale labels in the memory | true labels in the stored documents close the gap | older stored documents relabelled at each boundary$^{r}$ | 76.5±2.3 | 63.6±1.9 | partly |
| H2 | Current documents mislabel old entities | a loss that lets "O" mean "O or an old class" closes the gap | replay and current losses sliced to their classes | – | 60.5±10.1$^{\dagger}$ | open |
| H3 | The shared background normalisation | one background per session closes the gap without true labels, given a memory | CoLaR memory, session-blocked readout (CoLaR-SBR) | 89.1±2.0 | 88.4±2.6 | supports |
| | | | session-blocked readout, no memory | 11.4±2.3 | 21.6±2.2 | supports |
| H4 | Isolation replaces memory where a routing key exists | per-session deltas need no memory on one schema, and fail across schemas | per-session deltas, blocked readout, no memory (SBR-Recall) | 92.6±1.0 | 91.9±0.6 | supports |
| | | | the same on Mixed (AA), no memory | – | 44.7$^{\dagger,n=1}$ | supports |
| | | | the same on Mixed (AA), 50-document memory | – | 85.9$^{\dagger,n=1}$ | supports |
| | | | Joint training (reference) | 92.0±0.4 | 91.3±0.8 | |

pose it by cause, within label growth. Latent and exemplar-free replay, where forgetting sits by depth, continual relation extraction and the broader CL families: App. J.

## 3. Setup

**Streams and models.** A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token, scored by entity-level F1 (`seqeval`). **CIL-CORD**, the stream of every rung: CORD's 30 receipt classes arrive six at a time over five sessions; the head grows, and every entity outside the session, earlier or later, is labelled "O". **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema. **Mixed**: six sessions over three datasets and annotation schemas. LayoutLMv3-base [18] is the encoder of every rung; LiLT [50], BROS [16] and text-only BERT appear on DIL. Heads are linear. Table 2 summarises the streams; App. A gives their sessions.

Table 2. **Benchmark settings** (sessions: App. A). Train/test documents: FUNSD 149/50, SROIE 626/347, CORD 800/100. AA: mean F1 of the final model over the sessions; full-label F1: on the unmasked CORD test set (Sec. 3).

| | **CIL-CORD** | **DIL** | **Mixed** |
|---|---|---|---|
| Datasets | CORD | FUNSD, SROIE, CORD in turn | FUNSD, SROIE, CORD |
| Sessions | 5 | 3 | 6 |
| Label space | grows, 6 per session | fixed, unified | grows, 3 schemas |
| Classes (BIO tags) | 30 (61) | 4 (9) | 12 (25) |
| At a boundary | new classes | new domain | new classes, domain or schema |
| Reported | full-label F1, AA | AA | AA |

**Memory and comparison set.** Every memory-based rung uses one whole-document latent memory (CoLaR, Sec. 5.2): 50 documents per session stored as rank-128 SVDs of their layer-4 activations, replayed through the upper layers. The comparison set holds naive fine-tuning and joint training as references and one published method per family: ER [42],

DER++ [3], LwF [25], CODA-Prompt [46], CL-LoRA [15], LCA [48], and the continual-NER methods CPFD [56] and IS3 [39]; App. G gives their settings and our ports.

**Protocol.** AdamW, learning rate $5\times10^{-5}$, batch size 2, seeds 42, 7 and 123. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) and early-stops on 10% held out from its *training* data; the test split never selects a model. Cells marked † come from earlier runs (*legacy protocol*) that early-stop on the test split and vary in epoch cap and precision. A cell is clean only when all three clean seeds exist (App. A, G).

**Scoring.** The headline number is the *full-label F1*: the final model on the complete CORD test set with all 30 classes, as in continual NER [30]. Only clean runs record it; against a cell with only AA (†, DIL, Mixed) we compare in AA. Retention needs per-session test splits, whose labels show only that session's classes; from $R[i, j]$, the F1 on session $j$ after session $i$, AA $= \frac{1}{T}\sum_j R[T, j]$. Scored raw, such labels charge a model for every correct entity of another session, so joint training reads 33 (App. B); we score each session *session-restricted*, mapping a predicted tag outside session $j$'s label set to "O" before span matching. AA averages the sessions, while session 1 holds 57% of the 1,306 test spans; on every seed both metrics order the groups of clean runs the same way (App. B). Tuning and evaluating on the same scenario overstates CL methods [6]. The comparison set was fixed after a first grid on all three streams, the memory settings were chosen on DIL and carried unchanged, SBR's claim threshold (0.5) was not tuned, and the one setting chosen on CIL-CORD is SBR-Recall's delta capacity; App. G, F and D give the detail, including the methods dropped.

## 4. The investigation

Every rung below runs on CIL-CORD with LayoutLMv3 under the clean protocol and changes one thing against the run it is compared with, unless stated (H0's memory rows change several). Table 1 lists the rungs, and Fig. 2 draws them with where the final models' old-session errors go (as a table in App. B). Each rung states its prediction before its result, and a verdict word is one the numbers justify on every seed.

### 4.1. H0: is it just harder forgetting?

**Prediction.** If label growth only made drift harder, the memory that retains under domain shift would still retain here, a raw memory would do better, and the losses would look like naive forgetting, where old sessions are lost and the newest is learned well.

**Test and result: refuted.** The CoLaR memory comes within 2.2 AA points of joint training on DIL on every seed (means $87.8^{\dagger}$ against $88.7^{\dagger}$). On CIL-CORD with a shared softmax it reaches $30.7\pm1.5$ full-label F1 ($33.5\pm1.0$ AA) against $92.0\pm0.4$ ($91.3\pm0.8$). Raw documents do no better (ER, 24.3 $AA^{\dagger}$, below every CoLaR seed), but ER also trains every layer, keeps 200 documents and is a legacy run, so this row does not isolate memory fidelity. What separates the memory run from naive forgetting is the session being trained: its newest session ends at 2.0 to 16.0 F1, whereas naive training learns every session to at least 78.6 on every seed. The memory run also freezes layers 1..4 after the first session and replays four stored documents per step; with the same freezing and replay, the blocked readouts with and without the memory learn every session to at least 81.1 (App. D), so lost plasticity does not explain the suppression. It also keeps 18.7 to 22.6% of the old sessions' spans where naive training keeps none, at an old-span precision of at least 71.6% (a lower bound from the exact full-label false positives; App. B). The type of the lost entities does not separate the two runs. A span-level indicator (not a bound: on one blocked run it reads 81% where the per-token count is 0; App. B) puts 89 to 95% of CoLaR's missed spans entirely at "O", and at least 98% of naive training's: a shared softmax misses old entities at "O" with or without a memory.

**What it rules out, and what it leaves.** Drift alone does not explain the failure: the memory that answers drift on DIL leaves the session being trained suppressed here, which naive training, with all of its drift, does not show. Drift is still real: naive training keeps nothing of the old sessions, and on a fixed label space forgetting sits in the output-facing computation on all four encoders, as known for image classifiers [40]. App. F holds that evidence, a freezing experiment that moves drift rather than removing it (one seed, the mirror image of Kalb and Beyerer [19]), and replay against the classical remedies.

### 4.2. H1: are the stored labels stale?

**Prediction.** A document stored at session $t$ carries "O" on every entity of another session, earlier or later, so the memory itself calls background what the model must keep or learn. In detection the same staleness of replayed examples was named foreground shift and blamed for replay's failure [28]. If these stale labels are the cause, giving the stored documents their true labels should close the gap with the shared softmax unchanged.

**Test.** At the end of each session we give every document stored *before* it the true labels of all classes seen so far. The documents of the session just finished keep their own labels until the next boundary, and no refresh can cover the session trained next. The refresh rewrites 105/1019/1394

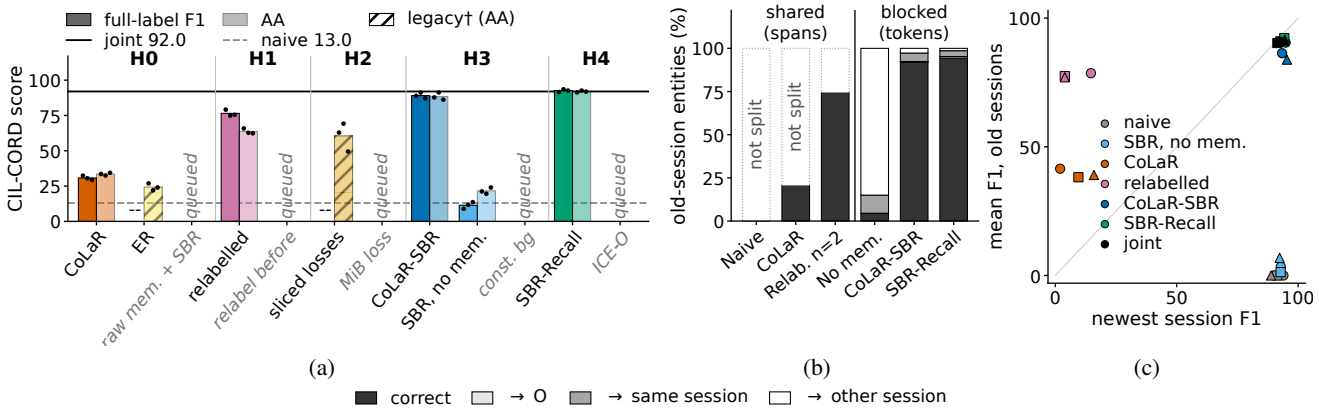



Figure 2. **The findings in one figure** (CIL-CORD, LayoutLMv3, clean protocol unless marked). (a) The ladder (Tab. 1): full-label F1 (dark) and AA (light); bar: mean of three seeds, dots: seeds; hatched: legacy (†), AA only; "–": no full-label run; lines: joint and naive full-label F1; "queued": not landed yet. (b) Where the final model puts old-session gold entities (sessions 1 to 4, mean over seeds); a shared softmax records only span recall. (c) Per seed (circle: 42, square: 7, triangle: 123), newest-session F1 against mean old-session F1: with a shared softmax and a memory the newest session is suppressed; naive training learns it and keeps nothing older. More pictures: App. I.

stale tokens before sessions 3 to 5 (and 1239 after the last session, which changes no training; seed 42). It reads labels no continual learner has and is a diagnostic, not a method.

**Result: partly.** Against the same memory without the refresh it lifts full-label F1 from 30.7 to 76.5 and AA from 33.5 to 63.6 (three seeds): $72to79\%$ and $48to56\%$ of the seed-matched gap to joint training (92.0, 91.3), and still below CoLaR-SBR (89.1, 88.4) under both metrics. The metrics disagree because of what the refresh cannot reach. While session $t$ trains, its own classes are "O" in every stored document, and sessions 3 to 5 are learned only to 2 to 37 F1 on their own diagonal; sessions 3 and 4 rise to 74 to 88 one session later, once their own documents are stored and the older ones carry their labels, and the last session, which has no later boundary, ends at 7.5. AA weights it as a fifth of the score; it holds $8\%$ of the test spans.

**What it rules out, and what it leaves.** Stale stored labels are a large part of the failure, and they act on the session being trained as much as on old ones: a session suppressed while current and recovered one boundary later is what stale stored labels predict (three seeds so far), and it marks the limit of this control. Whether a memory that is never stale closes the gap within a shared softmax is what relabelling every stored document *before* each session tests; that control leaves the current documents as the only stale source.

### 4.3. H2: do current documents mislabel old entities?

**Prediction.** A current document labels every entity of an earlier session "O", so even a memory with true labels is opposed in every batch by documents that call the same strings background. This is the second side of the conflict named in detection [35]. If this pull is the cause, a loss that lets a current "O" mean "O or an earlier class", as MiB does for segmentation [5] and marginal-likelihood taggers do offline [12], should close the gap within a shared softmax. Prior work is mixed on whether a separated readout does better: offline, marginalising over the tags a dataset does not annotate was at least as good as separate outputs per dataset [12], while in segmentation SSUL's separated sigmoids, with other changes, improved on MiB [7].

**Test.** The decisive control keeps the memory and the shared softmax and uses MiB's unbiased cross-entropy in both directions: a current "O" token matches "O" or any earlier-session class, a replayed one "O" or any class introduced after storage (no distillation). Its seeds are not complete; until they are, the evidence is the label-side repairs we ran earlier, all legacy runs (†, test-split early stopping, which favours them), compared across protocols with clean CoLaR.

**Evidence so far: open.** Restricting the replay loss to the classes alive at storage gives 39.4 AA (seed 42). Restricting the current loss as well, so that a current "O" token is trained only against "O" and the new classes (ER-ACE's asymmetric cross-entropy [4], applied to current "O" tokens only), lifts seed 42 to 62.9, both gate runs: the one same-protocol number specific to this rung. It changes the stored side (H1) and the current side at once; its three seeds and the pseudo-label variants are in App. E. No repair exceeds 69.3 AA on any seed, below every seed of CoLaR-SBR. These runs suggest that the current side matters and do not size it. In the relabelled-memory run old-session recall stays at 71.5

to 77.3%, against 88.4 to 92.6% for CoLaR-SBR (Fig. 2b): the current documents' pull and the stale labels the refresh cannot reach both remain, and refreshing every stored document before each session and the MiB-style loss separate them.

### 4.4. H3: is it the shared normalisation?

**Prediction.** If the conflict lives in the shared normalisation, one background logit trained by documents that disagree about what is background, then one background logit per session, normalised with that session's tags only, should close most of the gap with no true labels. This is SS-IL's diagnosis of a shared softmax [1] with a background added, and ICE-O's readout [27] without its frozen encoder. It needs the memory: separating the labels does not undo drift, so without a memory the same readout should fail, and fail the way drift does.

**Result: supported, on every seed.** With the memory the session-blocked readout (CoLaR-SBR, Sec. 5.1) reaches 89.1±2.0 full-label F1 and 88.4±2.6 AA, 0.6 to 5.3 full-label points below joint training per seed and above the relabelled memory on every seed both have, under both metrics. The owning block claims 95.4 to 98.3% of the old entity tokens, and most remaining errors (56.6 to 71.6% of the wrong tokens) give the token another tag of its own session, a B/I swap of the right type included. Without the memory the same readout scores 11.4±2.3 (21.6±2.2 AA), and its errors go elsewhere: not one old entity token is predicted "O", 83.8 to 93.5% of the wrong ones receive another session's class, and each old block claims 17.0 to 24.3% of the tokens of its own split that are not its entities, against 0.6 to 1.7% with the memory. Old blocks on a drifted trunk claim far more than their own entities; that is drift, and the memory reduces it. Figure 1 draws the conflict on one receipt (App. E).

**What goes against it.** The readout does nothing alone, and it is not free: on DIL, where there is nothing to separate, CoLaR-SBR is below CoLaR (AA 86.2† against 87.8†).

**What discriminates what.** Drift (H0) is told apart from the label hypotheses (H1 to H3) by the session being trained: under a shared softmax it is suppressed with a memory and learned without one, and the freezing that comes with the memory does not suppress it under the blocked readout; and by the same memory coming within 5.3 full-label points of joint training on every seed once the background is not shared. Within the label hypotheses, the relabelled memory measures part of the stored side (H1, three seeds so far); the current side (H2) and the shared normalisation (H3) are not yet separated, which the never-stale memory and the MiB-style loss decide together. The error type shows drift only within the blocked readout; under a shared softmax naive training and CoLaR both miss old entities at "O" by a span-level indicator, so it does not separate drift from conflict there (App. B).

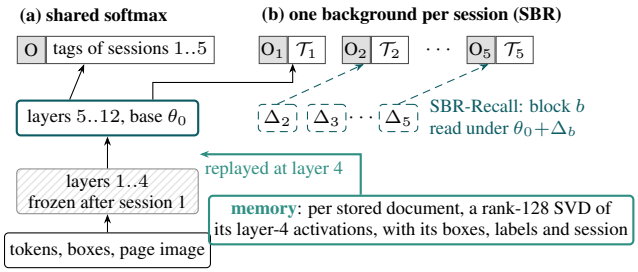



Figure 3. **The two readouts on one encoder and memory.** (a) One softmax over all tags and one background. (b) SBR: one background logit per session, trained only by documents that annotate it (Eq. (2)); the largest claim $e_b = 1 - p_b(\mathrm{O}_b) > 0.5$ labels the token. SBR-Recall drops the memory and reads block $b$ under its session's delta $\Delta_b$ (Eq. (3)).

## 5. From diagnosis to design

**The principle.** The rungs point at one fact about annotation under label growth, familiar from training on partially annotated data [12]. A token marked "O" in session $s$ means "none of the classes annotated in session $s$": it is evidence against those classes and silent about every other. A shared softmax turns it into evidence against all classes at once (H3), and stale stored labels are the same error made by the memory (H1). The principle: *supervise each class only with negatives from documents that annotated it, and normalise together only classes that were annotated together.* Three consequences follow: each session needs its own background; the memory must keep whole documents with their own labels and the session that wrote them; and each session's background logit, trained on that session's true negatives, is also a detector of its entities, so it can route a token without task identity.

### 5.1. SBR: one background per session

SBR is our instantiation of a per-session background readout [20, 27]. Every session that brings new tags opens a block $b$ holding those tags $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. SBR gives every session its own background logit $o_b(x) = w_b^\top h(x) + c_b$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\right), \qquad (1)$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block $b$ a token labelled "O" in session-$b$ data is truly background, so the block's supervision is never stale. For a token $x$ with label $y$

in a document of session $s$,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\ \mathrm{CE}\big(p_b(\cdot \mid x),\ y_b\big),$$
$$y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: $m_{sb}$=1 when session $s$ annotates every tag of $\mathcal{T}_b$; $m_{sb}$=1 with $y_b$=$\mathrm{O}_b$ when $b$ belongs to a different annotation schema (the document is annotated under its own schema, so every token is a negative for $b$; a training-time convention that, between related schemas such as SROIE and CORD, can call a real entity background); and $m_{sb}$=0 otherwise, because the document's "O" labels say nothing about block $b$. Stored documents follow the rule of the session that stored them. A session that adds no tags creates no block, so on a fixed label space SBR has a single block. At inference each block reports its claim $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest $e_b$ labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise, as in ICE-O's "Other unless some session scores above it". Blocks trained at different times need not produce claims on one scale, which is why we measure the routing directly (Fig. 2b; App. B). Figure 3 draws both readouts. App. C gives the pseudo-code, the supervision mode of every (session, block) pair and the memory accounting.

### 5.2. The memory

SBR's supervision needs stored documents that keep each token's label at its position and the session that wrote it. CoLaR (Compressed Latent Replay) keeps whole documents: after the first session layers $1..k$ are frozen, and each stored document's layer-$k$ activations $H_j$ are kept as a rank-$r$ truncated SVD with its boxes, mask, labels and session, rebuilt at replay and injected at layer $k$. On DIL it retains $87.8{\pm}0.5^{\dagger}$ AA in 60 MB, against $87.9{\pm}1.4^{\dagger}$ for ER with 200 raw documents; it holds no raw pixels or token ids (no privacy claim: activations can leak content). Summaries of the same store fall to the naive level (App. F).

### 5.3. SBR-Recall: the test of H4

**Prediction (H4).** With the conflict removed, the memory is needed only against drift, and the task-identity view [21] says when it can go: if each block is read under the trunk state that learned it, nothing drifts for that block, provided the model can tell which state a token belongs to. On one schema under label growth each block's background logit supplies that key. Across schemas the all-background terms of Eq. (2) train every old block towards background on other-schema documents with nothing to pull it back, so isolation should fail without stored documents; on a fixed label space there is one block and nothing to route by.

**The intervention.** SBR-Recall stores, for each session $b > 1$, a low-rank delta $\Delta_b$ [17] of the layers above $k$ (rank 64 on the attention projections and feed-forward layers), freezes these layers after the first session (the base $\theta_0$; $\Delta_1$=0), and computes block $b$ under the trunk state that learned it:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\,o_b,\ z_{\mathcal{T}_b}\,]\right)\big(x;\ \theta_0 + \Delta_b\big). \tag{3}$$

Every term of Eq. (2) is computed under the delta its block is read under, at training and test time. Without a memory on a single-schema stream no term of a later session involves an old block, so each old block's $e_b$ is unchanged up to weight decay, and only competition from new blocks can move its tokens (App. C).

**Result: supported, with its predicted failure.** On CIL-CORD SBR-Recall reaches $92.6{\pm}1.0$ full-label F1 ($91.9{\pm}0.6$ AA) with no stored document, within 2.1 points of joint training on every seed under both metrics; its owning block claims 97.0 to 98.3% of the old entity tokens. On Mixed without a memory it scores $44.7$ AA$^{\dagger}$ (seed 42): the blocks of the other schemas fall silent until their domain returns, as predicted, and with the 50-document memory restored it scores 85.9 (App. D). On a fixed label space SBR-Recall has a single block; that it cannot replace the memory there follows from the design and was not run. We ran no shared-softmax counterpart that combines or merges the deltas (as EASE and InfLoRA do); our CL-LoRA port shares one adapter across sessions.

## 6. Evidence beyond one stream

**The comparison set.** Table 3 reports the fixed comparison set. On CIL-CORD every continual method with a shared softmax stays at or below CoLaR in AA (33.5), and the two blocked readouts close most of the gap: CoLaR-SBR $89.1{\pm}2.0$ and SBR-Recall $92.6{\pm}1.0$ full-label F1 against $92.0{\pm}0.4$. Four baselines end below naive training on every seed because they learn later sessions poorly: CPFD learns no session after the first (at most 0.0 F1 on a later diagonal), and IS3, CODA-Prompt and LCA each learn some later session on every seed to at most 32.3 F1, where naive training learns every session to at least 78.6; like naive training they keep little of the earlier sessions (App. G). These rows are our ports onto one growing token head, not checked against their published NER numbers. On Mixed, where both causes act, CoLaR-SBR reaches 89.7 AA against 87.7 for joint training and 74.5 for ER. App. D lists the per-seed values behind every cell.

**Where the remaining gap lies.** With a shared softmax some sessions partly recover at a later boundary (Fig. 4); with the blocked readout every session is learned and one early session loses the most; under SBR-Recall the larger

Table 3. **Benchmark results on LayoutLMv3**, grouped by family. Memory: what a method keeps (sizes as recorded on CIL-CORD, SBR-Recall's deltas at two bytes per parameter). Readout: one shared softmax or one block per session. Mean ± s.d. over three seeds, per-seed values when fewer; "–": not run. $^{\dagger}$legacy protocol; $^{n=k}$: $k$ clean seeds so far; $^{c}$one fp16 LwF seed that collapsed to all-background is excluded; $^{b}$with the 50-document memory; $^{p}$made before we fixed our port's orthogonality penalty. Ports, hyper-parameters and protocol per cell: App. G.

| | | | | CIL-CORD | | Mixed | DIL |
|---|---|---|---|---|---|---|---|
| **Family** | **Method** | **Memory (what is stored)** | **Readout** | full-label F1 | AA | AA | AA |
| Reference | Naive fine-tuning | none | shared | 13.0±0.2 | 18.3±0.5 | 44.4/43.7$^{\dagger}$ | 41.3±1.5$^{\dagger}$ |
| | Joint training | all data, all sessions | shared | 92.0±0.4 | 91.3±0.8 | 86.9/88.5$^{\dagger}$ | 88.7±1.1$^{\dagger}$ |
| Regularisation | LwF (2016) | previous model | shared | – | 18.4/18.5$^{\dagger,c}$ | 47.7$^{\dagger,c}$ | 43.7±1.1$^{\dagger}$ |
| Replay | ER (2019) | 200 raw documents | shared | – | 24.3±2.4$^{\dagger}$ | 75.6/73.3$^{\dagger}$ | 87.9±1.4$^{\dagger}$ |
| | DER++ (2020) | 200 raw documents + logits | shared | – | 20.4±1.4$^{\dagger}$ | 75.1/74.0$^{\dagger}$ | 88.0±0.9$^{\dagger}$ |
| Prompt, LoRA | CODA-Prompt (2023) | none (prompt pool) | shared | – | 14.0/15.9$^{\dagger}$ | 12.1/10.6$^{\dagger}$ | 32.5±1.2$^{\dagger}$ |
| | CL-LoRA (2025) | none (one shared adapter) | shared | – | 18.7/18.6$^{\dagger}$ | 45.7/43.9$^{\dagger}$ | 41.1±0.3$^{\dagger,p}$ |
| Continual NER | CPFD (2023) | previous model | shared | – | 2.7/0.0$^{\dagger}$ | 0.3/15.1$^{\dagger}$ | 6.1±7.3$^{\dagger}$ |
| | IS3 (2024) | previous model | shared | – | 5.6/7.7$^{\dagger}$ | 0.0/40.6$^{\dagger}$ | 38.3$^{\dagger}$ |
| Merging | LCA (2026) | class Gaussians | shared | – | 13.7/7.0$^{\dagger}$ | 38.0/36.4$^{\dagger}$ | 43.9±2.5$^{\dagger}$ |
| This paper | CoLaR | 50 docs/session, rank-128 layer-4 latents (100.7 MB) | shared | 30.7±1.5 | 33.5±1.0 | 75.6/74.6$^{\dagger}$ | 87.8±0.5$^{\dagger}$ |
| | **CoLaR-SBR** | 50 docs/session, rank-128 layer-4 latents (100.7 MB) | blocked | 89.1±2.0 | 88.4±2.6 | **89.8**/**89.6**$^{\dagger}$ | 86.2±1.3$^{\dagger}$ |
| | + LexSlot head slots | + head slots, token ids | blocked | – | 89.7±0.5$^{\dagger}$ | 89.6±1.0$^{\dagger}$ | 87.0±0.4$^{\dagger}$ |
| | **SBR-Recall** | one rank-64 trunk delta per session (56.6 MB) | blocked, per delta | **92.6**±**1.0** | **91.9**±**0.6** | 85.9$^{\dagger,b}$ | – |

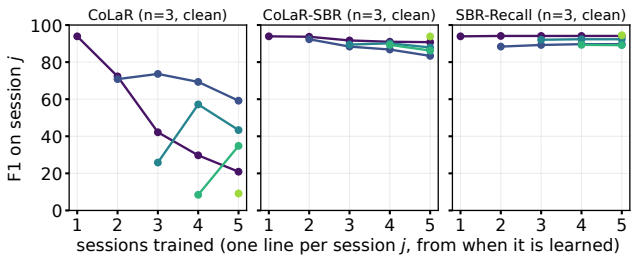



Figure 4. **Retention per session on CIL-CORD** (session-restricted F1 on session $j$ after each later session, mean over the seeds; panel titles give their number $n$ and the protocol, † = legacy). With a shared softmax the curves fall steeply and late sessions are learned poorly; with the blocked readout most sessions hold; under SBR-Recall no session falls more than 0.6 points below its diagonal on any seed.

moves, up to 2.1 points, are gains (App. D). Ablations and cost: App. C to E.

## 7. Limitations and conclusion

**Limitations.** The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; the ladder, SBR and SBR-Recall are evaluated on LayoutLMv3 and on one class-incremental partition of CORD. The freezing experiment, the memory-content ladder, the relabelled memory (three seeds so far) and SBR-Recall on Mixed are single- or few-seed diagnostics; H2 rests on legacy runs until its control lands. Errors are split per token only for the blocked readouts; for the shared softmax we have span counts and an indicator. We did not run several comparators from this literature (App. J); the MiB-style loss of Sec. 4.3 is the background-shift correction we queue on the same memory. Joint training is a reference, not a bound. SBR-Recall grows by one delta and one pass per session and freezes its base after session 1, so it may depend on the session order. SBR's rule reads each session's schema at training time. CoLaR's memory grows with the sessions, and storing no raw pixels or token ids is weaker than privacy.

**Conclusion.** Replay retains documents under domain shift and fails under label growth because two things go wrong, and a memory repairs only one. Drift is answered by a whole-document memory. The background conflict is not: stale stored labels are part of it, and relabelling the older stored documents with their true labels leaves a shared softmax below joint training and below CoLaR-SBR (three seeds); whether the current documents' labels are the rest is open. Under a shared softmax old entities are missed at "O" with or without a memory (by a span-level indicator), so there the error type does not tell the causes apart; what separates the conflict from drift is a suppressed current session, which appears only with a memory and a shared softmax. Most of the conflict is removed at the readout, by letting an "O" label count only against the classes its annotator could see, and where label growth supplies the routing key that readout lets per-session deltas replace the memory. Continual document methods should be tested on a growing label space, with a mask-free metric, against a memory whose stale labels are refreshed.

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